D.VISHNU
CSE
240701601

EXPERIMENT - 10

Aim

To create data visualizations such as pie charts and bar graphs for an inventory management system using JavaScript.

Introduction

Data visualization is used to represent data in graphical form, making it easier to understand and analyze. Charts like pie charts and bar graphs help in comparing values and identifying patterns.

In this experiment, pie charts and bar graphs are created to display inventory data using JavaScript and Chart.js.

Procedure :

Step 1: Create HTML File

index.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Inventory Visualization</title>

    <style>
        body {
            font-family: Arial;
            text-align: center;
        }

        .chart-container {
            width: 400px;
            height: 400px;
            margin: 20px auto;
        }
    </style>
</head>
<body>
<h2>Inventory Management System</h2>
<div class="chart-container">
    <canvas id="pieChart"></canvas>
</div>
<div class="chart-container">
    <canvas id="barChart"></canvas>
</div>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>

</body>
</html>
```

- Created an HTML file with title and headings
- Added two canvas elements for charts
- Linked Chart.js library

Step 2: Define Data in JavaScript

```js
const data = {
  labels: ['Electronics', 'Clothing', 'Books', 'Toys', 'Home Appliances', 'Cosmetics'],
  datasets: [{
    label: 'Items in Stock',
    data: [200, 150, 80, 50, 30, 230],
    backgroundColor: [
      'red', 'blue', 'green', 'orange', 'purple', 'cyan'
    ]
  }]
};
new Chart(document.getElementById('pieChart'), {
  type: 'pie',
  data: data,
  options: {
    responsive: true,
    maintainAspectRatio: false
  }
});
new Chart(document.getElementById('barChart'), {
  type: 'bar',
  data: data,
  options: {
    responsive: true,
    maintainAspectRatio: false,
    scales: {
      y: {
        beginAtZero: true
      }
    }
  }
});
```

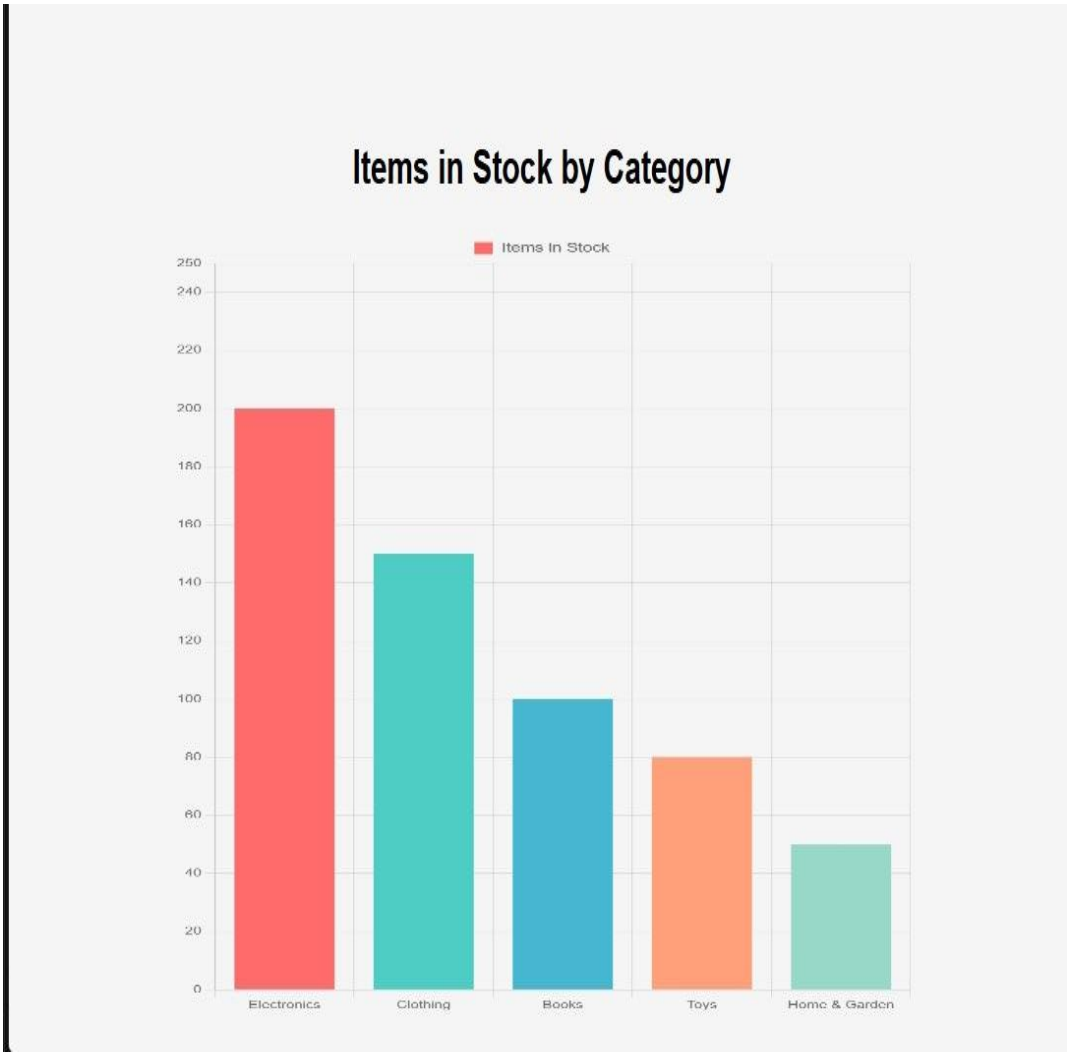
- Created data for categories like Electronics, Clothing, Books, etc.
- Assigned values for each category
- Defined labels and dataset

Step 3: Create Charts

- Used Chart.js to create:
    - Pie chart for distribution
    - Bar chart for comparison
- Configured chart type, labels, and colors

Step 4: Display Output

- Displayed charts on the webpage
- Verified correct visualization of data


Items in Stock by Category
Items In Stock
250
240
220
200
180
160
140
120
100
80
60
40
20
0
Electronics
Clothing
Books
Toys
Home & Garden